

EAS 501, Section 01, Fall 2026

Course title: Advanced Mathematical Methods - Advanced Math Methods

Instructor: Cheng Wang

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Office hours: Monday & Friday 3 to 4 pm, at the second floor of the library,
Wednesday, 3 to 5 pm, at Textile Building #001

Time and Place: MW 5:30 to 6:45 pm, at Textile Building #001

Textbook: *Advanced Engineering Mathematics*, Michael Greenberg, 2nd edition, 1998.

Course Description & Objectives: This is a graduate-level course on mathematical methods in science and engineering. In this course, we will cover: scalar and vector field theory, linear algebra, partial differential equations, and integral transforms.

The exams: There will be two regular in-class exams. Materials for these exams will be based on modified problems from Greenberg's book. To prepare for the exams, try your best to do example problems in the textbook, the assigned practice problems, and ask me questions during office hours if you need help.

The homework assignments: A few homework assignments will be given, and there will be a take-home project.

Lecture Schedule

Week 1 (09/02): 13.2, 13.3, 13.4 (Multivariate and Vector Calculus)

09/07, Labor Day, no class

Week 2 (09/09): 13.5, 13.6, 3.7, 1.38, 14.2, 14.3 (Multivariate and Vector Calculus)

Week 3 (09/14 and 09/16): 14.2, 14.3, 14.4, 14.5, 16.2, 16.3 (Multivariate and Vector Calculus)

Week 4 (09/21 and 09/23): 16.4, 16.5, 16.6, 16.7 (Multivariate and Vector Calculus, Linear Algebra)

Week 5 (09/28 and 09/30): 8.2, 8.3, Exam 1 (Linear Algebra)

Week 6 (10/05 and 10/07): 9.3, 9.4, 9.5, 9.6, 9.7, 9.8, 9.7, 9.8, 9.9, 9.10 (Linear Algebra)

10/12, Indigenous Day, no class 10/13, Following Monday's schedule

Week 7 (10/13 and 10/14): 10.2, 10.3, 10.4, 10.5 (Linear Algebra)

Week 8 (10/19 and 10/21): 10.6, 11.2, 11.3 (Linear Algebra)

10/28, Mental Health Day, no class

Week 9 (10/26): 11.4, 11.6

Week 10 (11/02 and 11/04): Exam 2, 17.2, 17.3, 17.4 (PDE and Integral Transforms)

11/11, Veterans Day, no class

Week 11 (11/09): 17.5, 17.7, 17.8 (PDE and Integral Transforms)

Week 12 (11/16 and 11/18): 17.9, 17.10, 5.2, 5.3, 5.4, 18.2, 18.3, 18.4 (PDE and Integral Transforms)

Week 13 (11/23 and 11/25): 19.2, 19.3, 19.4, 20.2, 20.3, 20.4 (PDE and Integral Transforms)

Thanksgiving: 11/26 to 11/29

Week 14 (11/30 and 12/02): 19.3, 19.4, 20.2, 20.3, 20.4 (PDE and Integral Transforms)

Week 15 (12/07 and 12/09): study and review

Grading:

The overall grade will be weighted as follows:

Homework: 20%, **Exam 1, 2:** 25% each, **Take-home project:** 30%.